Nonhomogeneous Equations & Undetermined Coefficients

ANSWER KEY

Complementary function and particular solution

1 For $y'' - 3y' + 2y = 4e^{3x}$, find the complementary function y_c.

$r^2 - 3r + 2 = (r-1)(r-2) = 0 \Rightarrow r = 1, 2$. So $y_c = C_1 e^x + C_2 e^{2x}$.

2 Explain why the general solution of a nonhomogeneous linear equation is $y = y_c + y_p$.

If y_p is any particular solution and y_c solves the associated homogeneous equation, then $y_c + y_p$ solves the nonhomogeneous equation (by linearity/superposition) and carries enough arbitrary constants to match any initial condition — so it is the general solution.

Choosing a trial form (no resonance)

3 Find a particular solution of $y'' - 3y' + 2y = 4x$.

Try $y_p = Ax + B$: $y_p' = A$, $y_p'' = 0$. Substituting: $-3A + 2(Ax+B) = 4x \Rightarrow 2Ax + (2B - 3A) = 4x$. So $2A = 4 \Rightarrow A = 2$, and $2B - 3A = 0 \Rightarrow B = 3$. Thus $y_p = 2x + 3$.

4 Find a particular solution of $y'' + y' - 2y = 6e^{3x}$ (note $r = 3$ is not a root of $r^2 + r - 2 = 0$).

Try $y_p = Ae^{3x}$: $y_p' = 3Ae^{3x}$, $y_p'' = 9Ae^{3x}$. Substituting: $9A + 3A - 2A = 6 \Rightarrow 10A = 6 \Rightarrow A = \frac{3}{5}$. Thus $y_p = \frac{3}{5}e^{3x}$.

The modification rule (resonance)

5 Find a particular solution of $y'' - 3y' + 2y = e^{2x}$. (Note $r = 2$ IS a root of $r^2 - 3r + 2$.)

Since e^{2x} solves the homogeneous equation, try $y_p = Axe^{2x}$. Then $y_p' = Ae^{2x} + 2Axe^{2x}$ and $y_p'' = 4Ae^{2x} + 4Axe^{2x}$. Substituting: $(4Ae^{2x} + 4Axe^{2x}) - 3(Ae^{2x} + 2Axe^{2x}) + 2Axe^{2x} = Ae^{2x}$ (the xe^{2x} terms cancel). Setting $Ae^{2x} = e^{2x}$ gives $A = 1$. Thus $y_p = xe^{2x}$.

Combining forcing terms

6 Find a particular solution of $y'' - y = e^x + x$. (Roots of $r^2 - 1 = 0$ are $r = \pm 1$, so e^x resonates but x does not.)

For the e^x part (resonant): try Axe^x, giving $y_{p1}'' - y_{p1} = 2Ae^x = e^x \Rightarrow A = \frac{1}{2}$, so $y_{p1} = \frac{1}{2}xe^x$. For the x part: try $Bx + C$, giving $-(Bx + C) = x \Rightarrow B = -1, C = 0$, so $y_{p2} = -x$. Total: $y_p = \frac{1}{2}xe^x - x$.

Application: a driven oscillator

7 A forced oscillator satisfies $y'' + 4y = \sin 3t$ (forcing frequency $3 \neq$ natural frequency 2 — no resonance). Find a particular solution.

Try $y_p = A\sin 3t + B\cos 3t$: $y_p'' = -9A\sin 3t - 9B\cos 3t$. Substituting: $-5A\sin 3t - 5B\cos 3t = \sin 3t \Rightarrow A = -\frac{1}{5}$, $B = 0$. Thus $y_p = -\frac{1}{5}\sin 3t$.

8 Now solve $y'' + 4y = \sin 2t$, where the forcing frequency EQUALS the natural frequency (resonance). Find a particular solution.

Since $\sin 2t, \cos 2t$ solve the homogeneous equation, try $y_p = t(A\sin 2t + B\cos 2t)$. Computing y_p'' and substituting, the terms proportional to $t\sin 2t$ and $t\cos 2t$ cancel, leaving $y_p'' + 4y_p = 4A\cos 2t - 4B\sin 2t$. Setting this equal to $\sin 2t$ gives $A = 0$, $B = -\frac{1}{4}$. Thus $y_p = -\frac{1}{4}t\cos 2t$ — the amplitude grows linearly in t, the signature of resonance.

More practice: no resonance

9 Find a particular solution of $y'' - 4y = 8x^2$.

Try $y_p = Ax^2 + Bx + C$: $y_p'' = 2A$. Substituting: $2A - 4(Ax^2 + Bx + C) = 8x^2 \Rightarrow -4A = 8, -4B = 0, 2A - 4C = 0$. So $A = -2$, $B = 0$, $C = -1$. Thus $y_p = -2x^2 - 1$.

10 Find a particular solution of $y'' + 5y' + 6y = e^x$ (roots of $r^2 + 5r + 6$ are $-2, -3$; no resonance).

Try $y_p = Ae^x$: $A + 5A + 6A = 12A = 1 \Rightarrow A = \dfrac{1}{12}$. Thus $y_p = \dfrac{1}{12}e^x$.

More practice: with resonance

11 Find a particular solution of $y'' - 4y' + 4y = e^{2x}$ (note $r = 2$ is a REPEATED root).

Since e^{2x} and xe^{2x} both solve the homogeneous equation, try $y_p = Ax^2e^{2x}$. Substituting and simplifying, the x and x^2 terms cancel appropriately, leaving $2Ae^{2x} = e^{2x} \Rightarrow A = \dfrac{1}{2}$. Thus $y_p = \dfrac{1}{2}x^2e^{2x}$.

12 Find a particular solution of $y'' - y' = 3$ (note $r = 0$ IS a root of $r^2 - r = r(r-1)$, so constant forcing resonates).

Since a constant solves the homogeneous equation, try $y_p = Ax$ instead of a bare constant: $y_p' = A$, $y_p'' = 0$. Substituting: $0 - A = 3 \Rightarrow A = -3$. Thus $y_p = -3x$.

Combining a polynomial and an exponential forcing term

13 Find a particular solution of $y'' - 3y' + 2y = 2x + e^x$ (roots are $1, 2$; e^x resonates, $2x$ does not).

Polynomial part: try $Ax + B$: $-3A + 2(Ax + B) = 2x \Rightarrow A = 1$, $B = \dfrac{3}{2}$. Exponential part (resonant): try Cxe^x; substituting gives $-Ce^x = e^x \Rightarrow C = -1$. Total: $y_p = x + \dfrac{3}{2} - xe^x$.

Setting up the trial form (without solving)

14 For $y'' + 4y' + 3y = 5\cos 2x$, write the correct trial form for y_p (roots of $r^2 + 4r + 3$ are $-1, -3$; no resonance with sinusoidal forcing).

$y_p = A\cos 2x + B\sin 2x$.

15 For $y'' - 2y' + y = xe^x$ (note $r = 1$ is a REPEATED root, and the forcing already matches that root's exponential), write the correct trial form.

Since e^x and xe^x both solve the homogeneous equation (double root $r = 1$), the forcing xe^x needs an extra factor of x^2: $y_p = (Ax^2 + Bx^3)e^x$.

Application: a driven oscillator with polynomial forcing

16 A forced spring satisfies $y'' + 4y = t^2$ (no resonance, since the roots $\pm 2i$ are not 0). Find a particular solution.

Try $y_p = At^2 + Bt + C$: $y_p'' = 2A$. Substituting: $2A + 4(At^2 + Bt + C) = t^2 \Rightarrow A = \frac{1}{4}$, $B = 0$, $C = -\frac{1}{8}$. Thus $y_p = \frac{1}{4}t^2 - \frac{1}{8}$.

One more round of undetermined coefficients

17 Find a particular solution of $y'' + y = 3\cos x$ (roots of $r^2 + 1$ are $\pm i$, so $\cos x$ resonates).

Since $\cos x, \sin x$ solve the homogeneous equation, try $y_p = x(A\cos x + B\sin x)$. Substituting and simplifying (the $x\cos x$, $x\sin x$ terms cancel automatically), only $-2A\sin x + 2B\cos x$ survives, so $-2A = 0$ and $2B = 3$, giving $A = 0$, $B = \frac{3}{2}$. Thus $y_p = \frac{3}{2}x\sin x$.

18 Find a particular solution of $y'' - 2y' - 3y = e^{4x}$ (roots of $r^2 - 2r - 3$ are $3, -1$; no resonance).

Try $y_p = Ae^{4x}$: $y_p' = 4Ae^{4x}$, $y_p'' = 16Ae^{4x}$. Substituting: $16A - 8A - 3A = 5A = 1 \Rightarrow A = \frac{1}{5}$. Thus $y_p = \frac{1}{5}e^{4x}$.

One final trial-form problem

19 Find a particular solution of $y'' + 3y' = 9x$ (note $r = 0$ IS a root of $r^2 + 3r = r(r+3)$, so a bare polynomial resonates).

Since a constant solves the homogeneous equation, try $y_p = x(Ax + B) = Ax^2 + Bx$ instead of a plain $Ax + B$. Then $y_p' = 2Ax + B$, $y_p'' = 2A$. Substituting: $2A + 3(2Ax + B) = 9x \Rightarrow 6Ax + (2A + 3B) = 9x$. So $A = \frac{3}{2}$, and $2A + 3B = 0 \Rightarrow B = -1$. Thus $y_p = \frac{3}{2}x^2 - x$.

20 Find a particular solution of $y'' + y' - 2y = 10\sin x$ (roots of $r^2 + r - 2$ are $1, -2$; no resonance with sinusoidal forcing).

Try $y_p = A\sin x + B\cos x$: $y_p' = A\cos x - B\sin x$, $y_p'' = -A\sin x - B\cos x$. Substituting: $(-A\sin x - B\cos x) + (A\cos x - B\sin x) - 2(A\sin x + B\cos x) = 10\sin x$, i.e. $(-3A - B)\sin x + (A - 3B)\cos x = 10\sin x$. So $-3A - B = 10$ and $A - 3B = 0 \Rightarrow A = 3B$. Substituting: $-9B - B = 10 \Rightarrow B = -1$, $A = -3$. Thus $y_p = -3\sin x - \cos x$.

A closing pair

21 Find a particular solution of $y'' - y = 4$.

Try $y_p = A$ (constant): $y_p'' = 0$. Substituting: $0 - A = 4 \Rightarrow A = -4$. Thus $y_p = -4$.

22 Find a particular solution of $y'' + 4y' + 4y = e^{-2x}$ (note $r = -2$ is a REPEATED root).

Since e^{-2x} and xe^{-2x} both solve the homogeneous equation, try $y_p = Ax^2e^{-2x}$. Substituting and simplifying leaves $2Ae^{-2x} = e^{-2x} \Rightarrow A = \frac{1}{2}$. Thus $y_p = \frac{1}{2}x^2e^{-2x}$.

A final full worked example

23 Solve the initial value problem $y'' - y = 2e^x$, $y(0) = 1$, $y'(0) = 0$ (roots of $r^2 - 1$ are ± 1, so e^x resonates).

$y_c = C_1e^x + C_2e^{-x}$. For y_p, try Axe^x: substituting gives $2Ae^x = 2e^x \Rightarrow A = 1$, so $y_p = xe^x$. General solution: $y = C_1e^x + C_2e^{-x} + xe^x$. From $y(0) = 1$: $C_1 + C_2 = 1$. $y' = C_1e^x - C_2e^{-x} + e^x + xe^x$; $y'(0) = C_1 - C_2 + 1 = 0 \Rightarrow C_1 - C_2 = -1$. Solving: $C_1 = 0$, $C_2 = 1$. Thus $y = e^{-x} + xe^x$.